

Part 2 – Python Traffic Analytics Pipeline

Methodology and Findings Report

1. Methodology

Part 2 extended the Part 1 traffic analysis into a reproducible Python workflow using Pandas, NumPy and Matplotlib. The workflow covered data cleaning, feature engineering, visualisation and a command-line analytics application. Python logging provided an auditable record of processing events, warnings and errors.

2. Data Cleaning

The raw dataset contained 48,204 rows and 9 columns. The pipeline validated the expected schema before processing, parsed and validated date/time values, standardised categorical values and identified duplicate and anomalous records. It removed 17 duplicate rows, handled 10 physically implausible 0 Kelvin temperature readings and 120 rainfall values outside the defined plausible range. Affected numeric values were imputed using monthly medians. The cleaned dataset contained 48,187 observations.

3. Feature Engineering

The cleaned data was transformed into an ML-ready dataset containing 36 columns. Features included hour, day of week, weekend status and cyclical time encodings. Weather categories were one-hot encoded, with additional precipitation, severe-weather, low-visibility and holiday indicators. Temperature and traffic volume were scaled. A data-driven congestion category used quartiles of traffic volume (Q1 = 1,192.5; Q2 = 3,379; Q3 = 4,933).

4. Findings

The hourly analysis showed a strong time-of-day pattern: average traffic was lowest at about 3:00 AM (371 vehicles/hour) and highest at about 4:00 PM (5,664 vehicles/hour). Weekday traffic averaged about 3,533 vehicles/hour versus 2,571 on weekends. The cleaned temperature-versus-traffic correlation was approximately 0.132, indicating a weak positive relationship.

5. Mini Application and Reproducibility

The CLI supports hourly traffic queries, high-traffic periods, weekday/weekend comparison, lower-traffic recommendations, weather queries and date-specific information. Invalid inputs are logged as errors rather than producing uncontrolled tracebacks. Separate modules, generated figures, processed datasets, a shared pipeline log, README and requirements file support reproducible execution.

6. Smart-City Implications

The results indicate that temporal patterns are more informative for operational traffic management than temperature alone. Monitoring, commuter information and demand-management activities can therefore focus on predictable peak periods, while weather is used as supporting context. The cleaned and engineered dataset provides a stronger foundation for predictive modelling and MLOps work in Part 3.